# Parth Bhosale

+91-9035887794 | parthbhosale35@gmail.com | github.com/StellarYolk | LinkedIn

## Education

**Indian Institute of Technology, Bhilai (IIT-BH)**
*Bachelor of Technology, Computer Science and Engineering* — *Jul 2024 – Present*

## Experience

**Jr. Software Developer Intern** — Bengaluru, Karnataka
*KSR Datavizon* — *May 2026 – Jul 2026*

- Built four AI-driven features end-to-end – an AI interview system, resume analyzer, mentor-matching chatbot, and mentor CRM dashboard – from loosely scoped specs, handling model selection through deployment under real cost and hardware constraints
- Re-engineered the speech-to-text pipeline for the AI interview system, cutting transcription cost by 28% and response latency by 0.5s while improving accuracy on domain-specific vocabulary
- Built a resume analysis feature that scores resumes against a target role (1–10) and generates actionable feedback on content, phrasing, and relevance, with pre-filled resume templates
- Built a mentor-matching chatbot for professionalhire.com ($\sim$100 mentors): filtered DynamoDB mentor records by skill/experience requirements extracted from a resume or query, then used an LLM to rank and return the best match
- Built a real-time CRM dashboard for mentors showing session history and derived analytics – peak booking times, earnings, and repeat-candidate tracking

## Projects

**Microparticle Detection & Navigation System** | *Ongoing Research, Patent Pending* — Feb 2026 – Present

- Patent submitted and under review, research papers for the same under review
- Trained and deployed YOLOv8n model using Python for microscale particle detection, augmented with a custom post-processing filter using geometric descriptors (aspect ratio, solidity, extent) to suppress false positives from visually similar particles
- Designed a hybrid tracker switching from YOLO detection to intensity-thresholded blob tracking post-selection, improving robustness without additional inference overhead
- Developed a ring-particle discrimination algorithm exploiting interior intensity differences between microbots and background particles, eliminating false locks without retraining
- Built an adaptive ROI system that dynamically resizes the detection window based on tracking confidence, reducing per-frame inference area and improving throughput
- Currently working on guiding microbot through microscopic maze, detected using OpenCV, using bi-directional BFS algorithm

**Email Classifier** | github.com/StellarYolk/Email-Classifier

- Built a hand-rolled agentic pipeline (no agent framework) that classifies unread Gmail with a locally-run LLM (Ollama, Llama3.2:3b) and auto-sorts it into Gmail labels via a classify $\rightarrow$ validate $\rightarrow$ act $\rightarrow$ record loop
- Constrained LLM output to a strict JSON schema with Pydantic, with automatic retry on validation failure and fallback to a held/unknown state for human review
- Designed a confidence-aware routing system – auto-sorted and auto-archived labels apply immediately, while ambiguous cases are queued for an interactive user decision after each batch
- Implemented resumable batch processing via a cursor file, so an interrupted run skips already-processed emails and continues cleanly

## Technical Skills

**Programming**: C++, Python, C, Shell Scripting, JavaScript, MySQL, PostgreSQL
**Libraries/Frameworks**: LangChain, LangGraph, Hugging Face, Pydantic, Scikit-Learn, PyTorch, OpenCV, Ultralytics (YOLO), MediaPipe, Ollama, NumPy, Pandas, Vite, Matplotlib, FastAPI, MongoDB, Express.js, React.js, Node.js, Uvicorn
**Tools & OS**: Git, GitHub, Docker, Linux, OpenCode, OAuth